EscapeAssist - Software Requirements Specification (SRS)

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Project Description: EscapeAssist is an AI-powered retrieval-augmented assistant designed to provide accurate, grounded answers from the 2022 Ford Escape Owner's Manual. The system includes a custom preprocessing pipeline, a multi-agent evaluation harness, and a Streamlit UI with analytics dashboards.

Section 1: Introduction

Introduction: EscapeAssist was chosen because owner's manuals are notoriously difficult to search, especially for new or inexperienced vehicle owners. As a 2022 Ford Escape Hybrid owner, the developer has firsthand experience with the challenges of navigating a 500-page manual. This project merges personal passion with modern AI engineering, creating a meaningful and technically challenging capstone.

Purpose: EscapeAssist will allow Ford Escape owners to ask natural-language questions and receive accurate, grounded answers sourced directly from the 2022 Owner's Manual. The system will improve accessibility, reduce frustration, and provide a safer, more intuitive way to access critical vehicle information.

Scope

- EscapeAssist will retrieve accurate information from the 2022 Ford Escape manual using a preprocessing-optimized RAG pipeline.
- EscapeAssist will provide a Streamlit UI for conversational interaction.
- EscapeAssist will include a golden question dataset and automated evaluation harness.
- EscapeAssist will produce accuracy, hallucination, and category-level metrics.
- EscapeAssist will display results in a dashboard.

Out of Scope: EscapeAssist will not provide repair instructions beyond what is in the manual, diagnose real-time vehicle issues, or support all model years. Only the 2022 model year is required, and support for other years is optional.

Technologies Used

- Python
- Streamlit
- OpenAI Platform (Agents, Embeddings, Judge LLM)
- GitHub Copilot
- PDF preprocessing libraries
- Custom evaluation harness

Section 2a: Must-Have Requirements

1. The system shall ingest and preprocess the 2022 Ford Escape Owner's Manual.
2. The system shall retrieve grounded answers using a RAG pipeline.
3. The system shall provide a Streamlit UI for user interaction.
4. The system shall maintain chat history within a session.

5. The system shall include a golden question dataset covering major categories.
6. The system shall run an automated evaluation harness using a judge LLM.
7. The system shall compute accuracy and hallucination metrics.
8. The system shall display evaluation results in a dashboard.
9. The system shall allow end-to-end demonstration of retrieval, evaluation, and analytics.

Section 2b: Stretch Requirements

1. The system may provide additional dashboard visualizations (trend lines, confusion matrices, error breakdowns).
2. The system may export evaluation results to CSV or JSON.
3. The system may enable users to run evaluations directly from the dashboard with a progress indicator.
4. The system may provide more interactive visualizations or filters in the evaluation dashboard to help users explore results by category, date, or agent version.
5. The system may enable export of evaluation results in multiple formats (e.g., CSV, JSON) with customizable fields.
6. The system may provide summary statistics or trend insights over time if multiple evaluation runs are stored.

Section 2c: Weekly Schedule

- Week 1: Research & Proposal
- Week 2: Foundry Setup & Proposal Completion
- Week 3: Document Preprocessing
- Week 4: Prompt Engineering & Retrieval Validation
- Week 5: Streamlit UI Prototype
- Week 6: Evaluation Harness Design
- Week 7: Judge LLM Integration
- Week 8: Dashboard Foundations
- Week 9: Dashboard Completion
- Week 10: Error Analysis & Iteration
- Week 11: Retrieval Enhancements
- Week 12: System Testing
- Week 13: Documentation
- Week 14: Final Project Reflection

Section 3: Design Overview of the Product

Workflow

1. User submits a question in the Streamlit UI.
2. Backend sends query to EscapeAssist agent.
3. Agent retrieves relevant manual chunks.
4. LLM generates a grounded answer.
5. Evaluation harness (offline) tests the agent using the golden dataset.
6. Dashboard visualizes metrics.

Resources

- 2022 Ford Escape Owner's Manual (PDF)
- Preprocessed manual versions

- Golden question dataset
- Evaluation results JSON

Data at Rest

- Preprocessed manual chunks
- Evaluation results
- Golden dataset
- Prompt templates

Data on the Wire

- User queries
- Retrieved context
- LLM responses
- Judge LLM scoring outputs

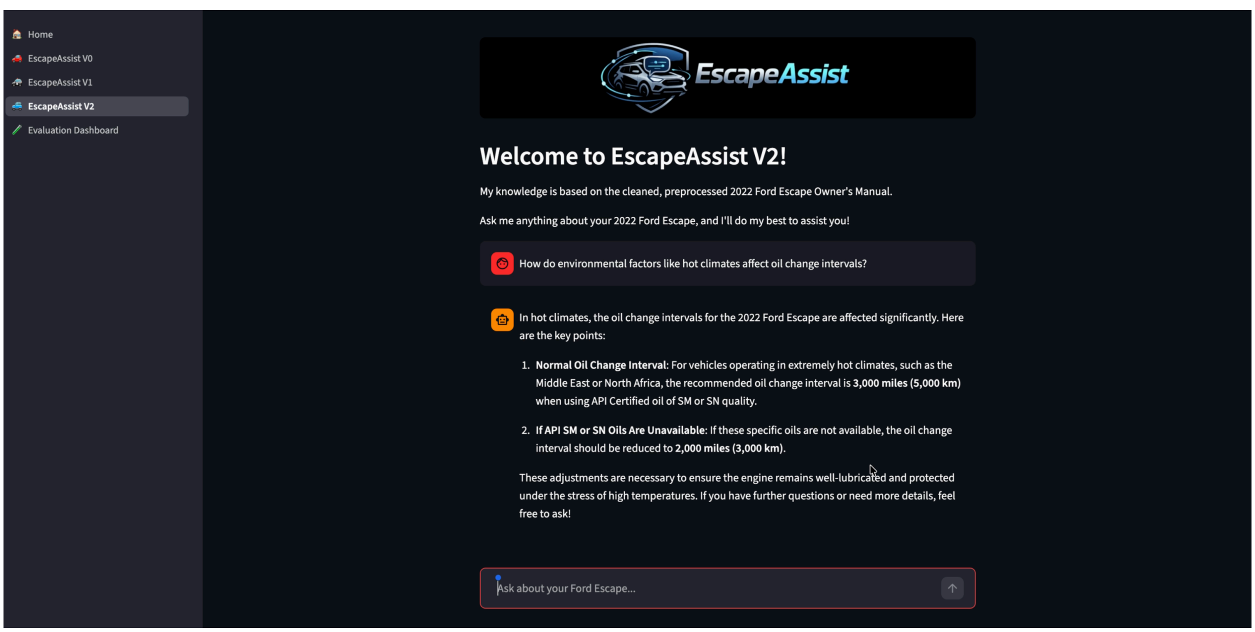
Data State

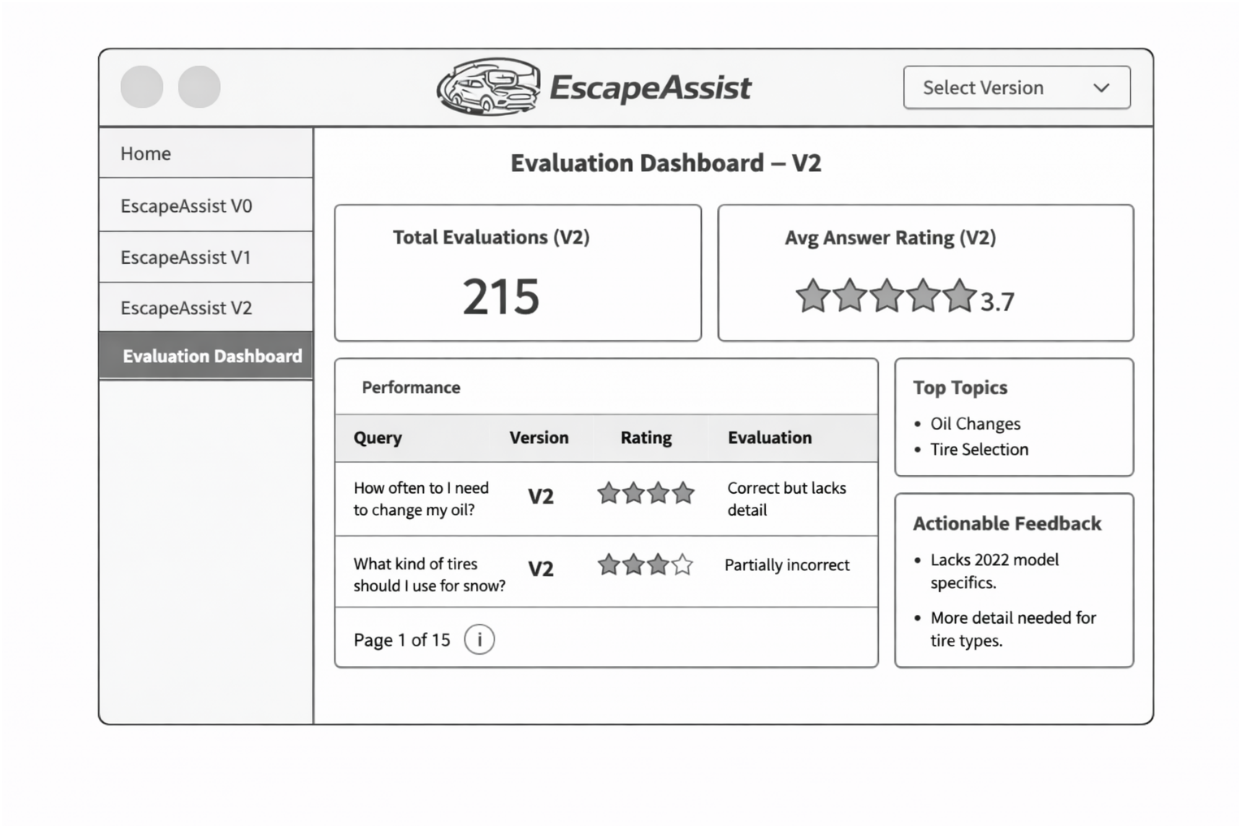
- Stateless API calls for agent queries
- Stateful session for UI chat history
- Persistent storage for evaluation results

HMI/HCI/GUI

- Streamlit chat interface
- Dashboard with charts, tables, and filters
- Evaluation summary views

Pictures/Diagrams

- 



Section 4: Verification

Demo

- Live demonstration of the Streamlit UI querying the EscapeAssist agent.
- Show retrieval grounding and answer accuracy.
- Demonstrate dashboard metrics.

Testing

- Golden dataset evaluation.
- Judge LLM scoring.
- Manual spot-checking of responses.
- Comparison of multiple preprocessing strategies.

Sources/Citations/Resource Links

- 2022 Ford Escape Owner’s Manual (Ford Motor Company)
- OpenAI Platform Documentation
- Streamlit Documentation
- GitHub Repository: https://github.com/NWilkins95/EscapeAssist